Project 1 — SafeLine Web Application Firewall (WAF)

Abstract: This project focuses on web application security using DVWA (Damn Vulnerable Web Application) and the SafeLine Web Application Firewall (WAF). DVWA, an intentionally vulnerable website, is deployed with Docker on an Ubuntu virtual machine and attacked with SQL injection to show how such attacks work. SafeLine WAF is then configured to protect the site, so the same attack is detected and blocked. Rate limiting is also configured to protect the application from abnormal request volumes from individual clients. The project teaches how web attacks work, how a WAF protects applications, and how security logs are used to monitor and analyze malicious activity.

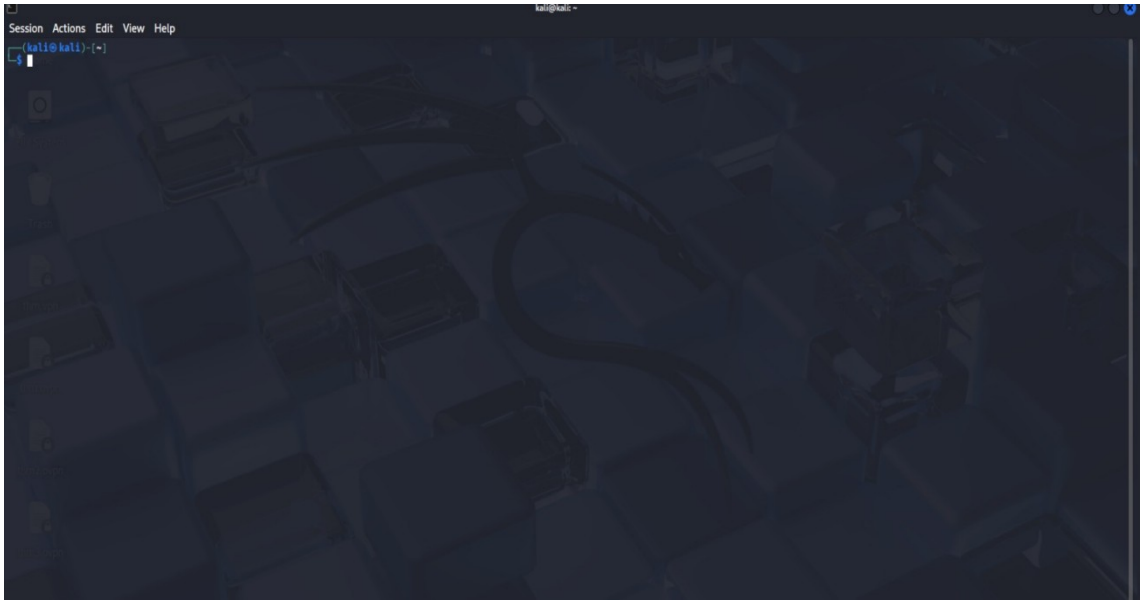
Requirements:

Software	VirtualBox (free) + Ubuntu 22.04 ISO (free) + Docker (free)
RAM	At least 4 GB allocated to the Ubuntu VM
Disk Space	20 GB free on your computer
Network	Set VM network to 'Host-Only Adapter' in VirtualBox settings

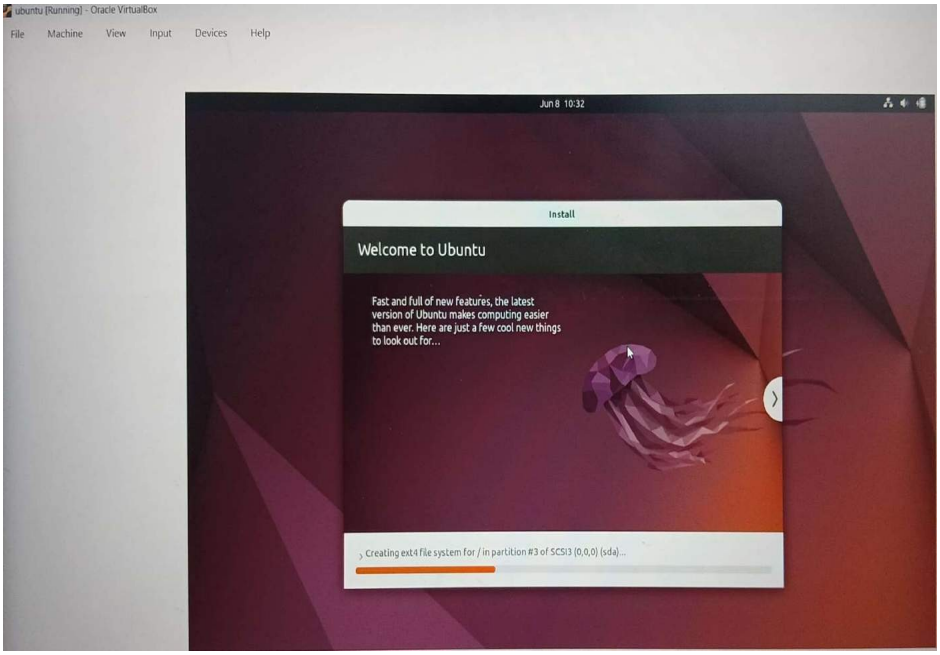
Step-by-Step Instructions

Phase 1 — Set Up Your Ubuntu Virtual Machine

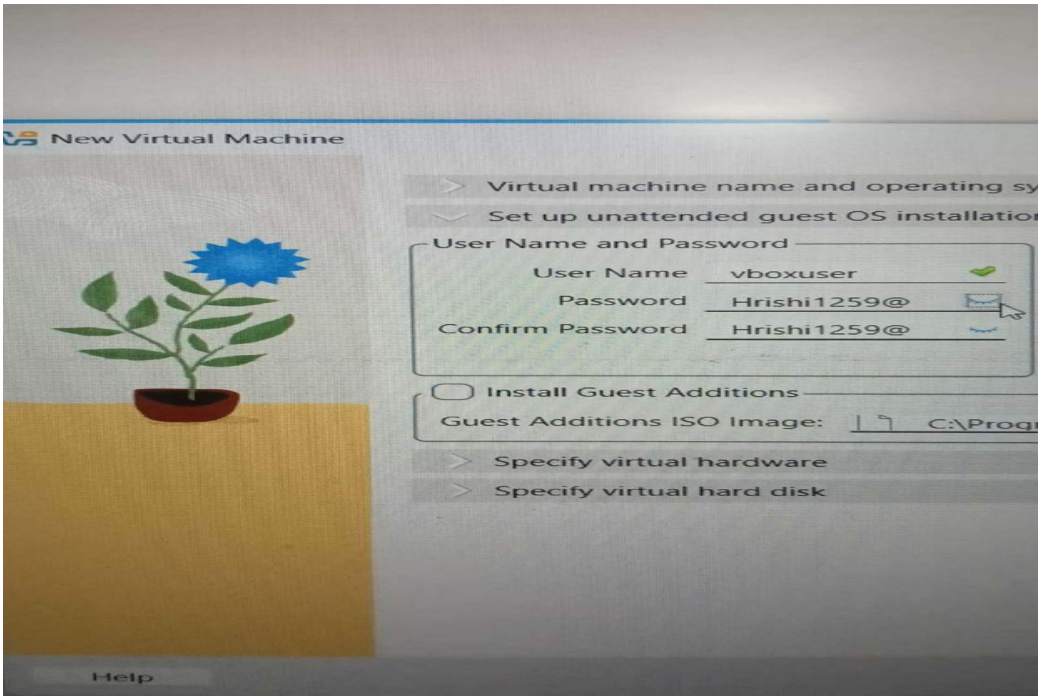
Step 1: Download VirtualBox — Go to virtualbox.org and download the installer for your operating system. Install it like any normal program.



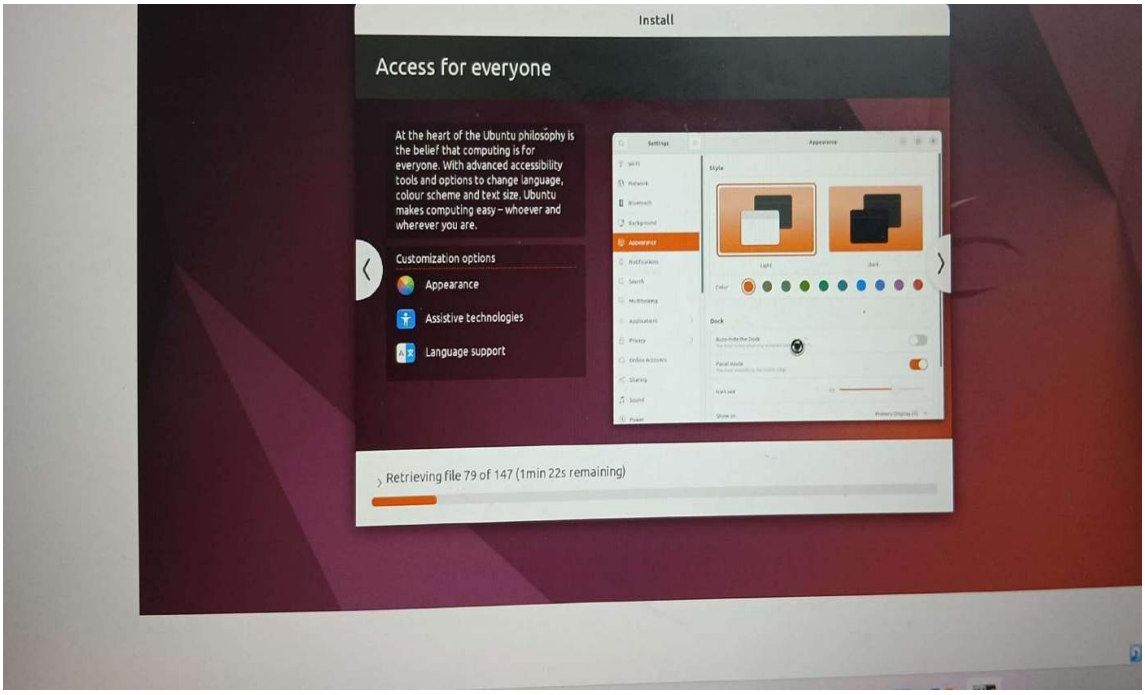
Step 2: Download Ubuntu — Go to ubuntu.com/download/desktop and download Ubuntu 22.04 LTS (the file is about 4–5 GB).



Step 3: Create a new VM — Open VirtualBox, click 'New', and name the VM 'WAF-Lab'. Set the type to Linux and the version to Ubuntu (64-bit). Allocate 4096 MB of RAM and create a 20 GB virtual hard disk.



Step 4: Attach the Ubuntu ISO — In the VM's Settings > Storage, click the empty disc icon and attach the Ubuntu ISO you downloaded. Start the VM and follow the Ubuntu installation wizard.



Step 5: Configure the network — In VirtualBox Settings > Network, set Adapter 1 to 'Host-Only Adapter'. This keeps your lab isolated from your home network.

Phase 2 — Install Docker

Docker is a tool that lets you run applications in containers — small, self-contained environments. DVWA and SafeLine both run as Docker containers.

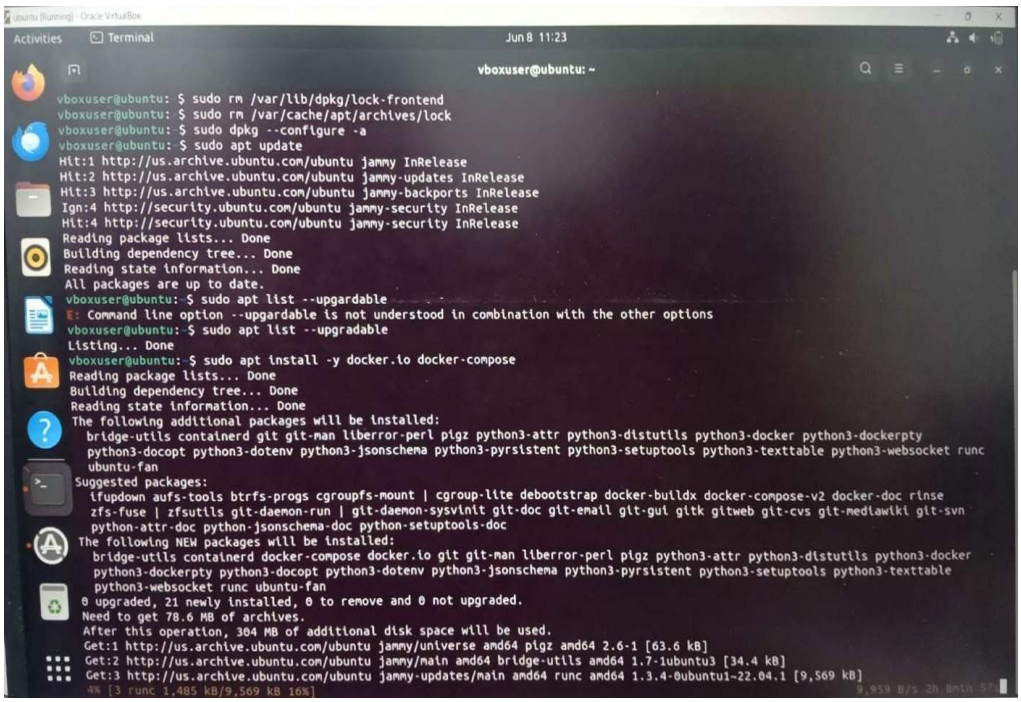
In the Ubuntu terminal, type these commands one at a time, pressing Enter after each:

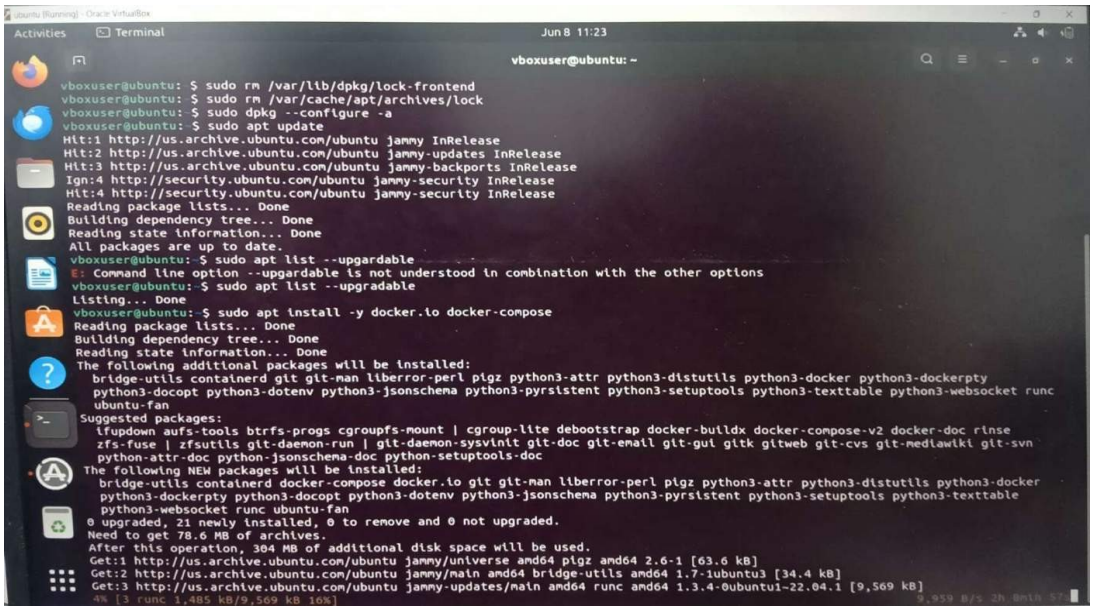
sudo apt update

sudo apt install -y docker.io docker-compose

sudo systemctl enable docker

sudo systemctl start docker

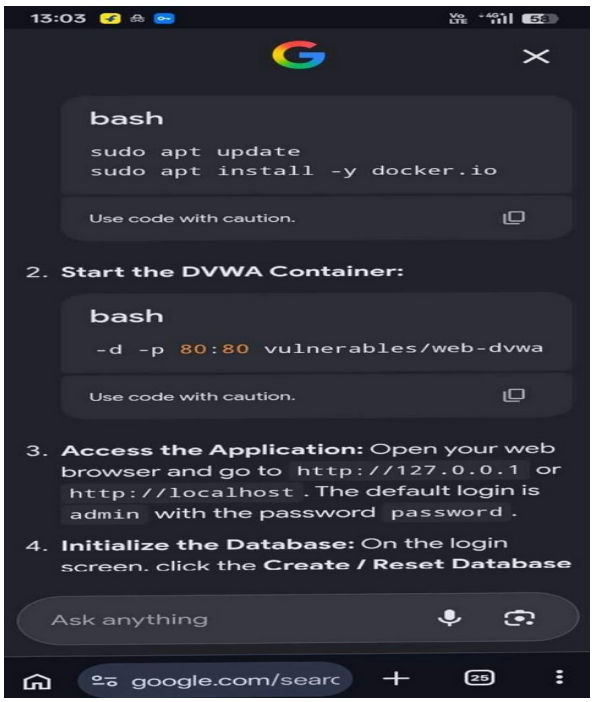


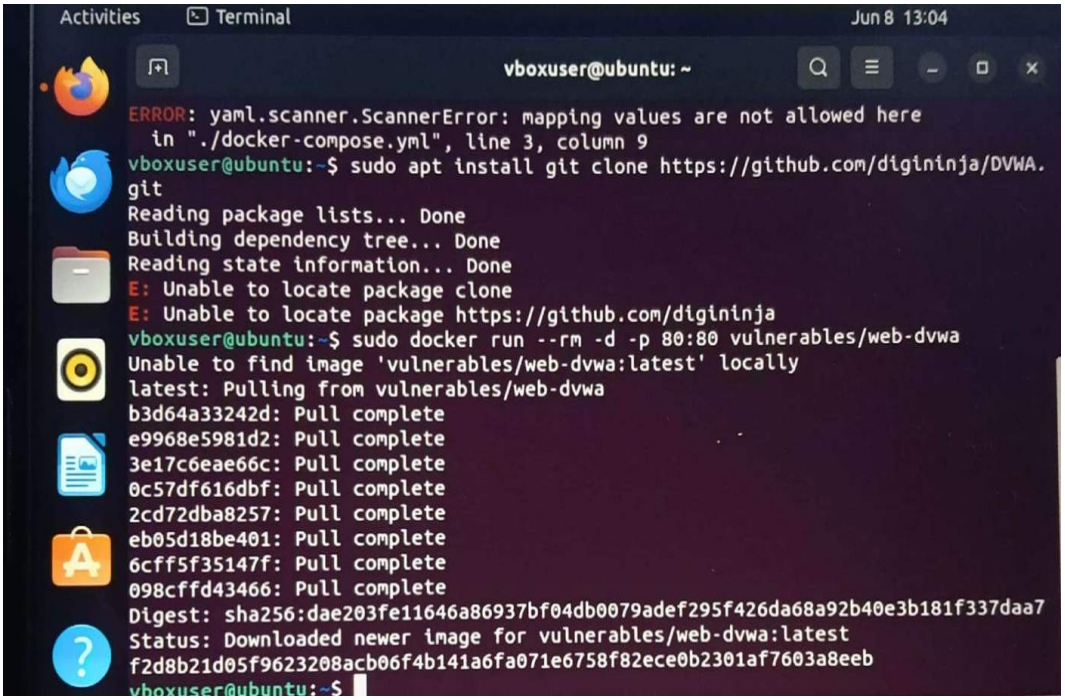


Phase 3 — Install DVWA (The Vulnerable Website)

Step 6: Create a folder — In the terminal, type: mkdir dvwa-lab && cd dvwa-lab

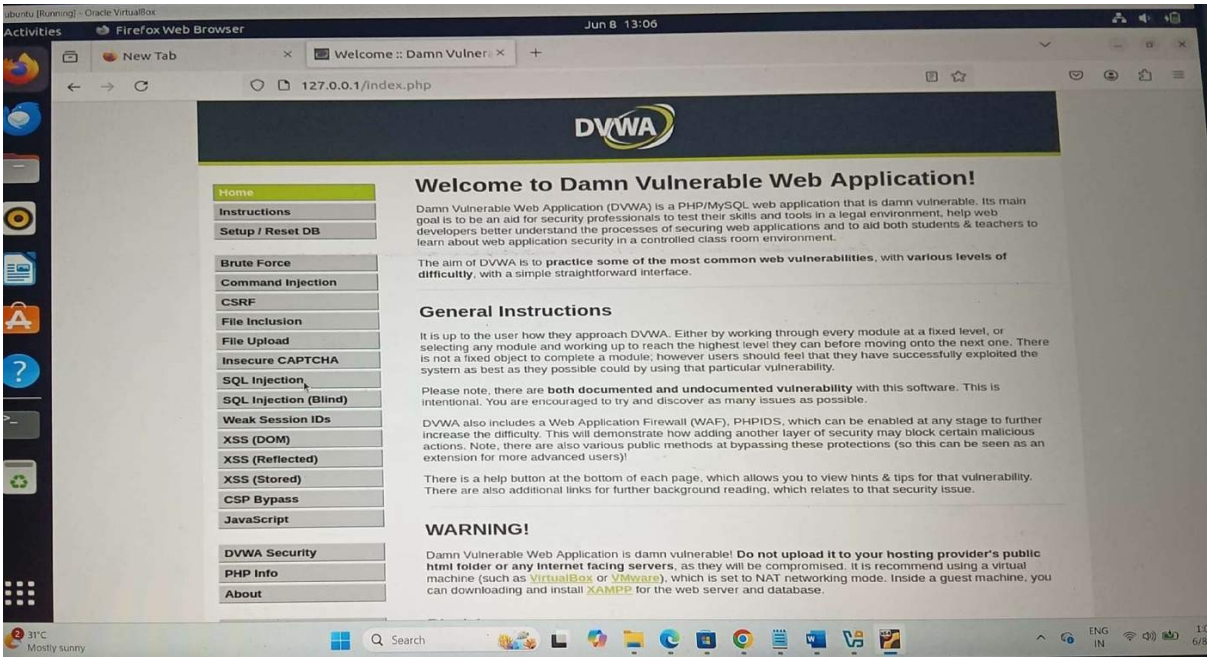
Quick option — start DVWA directly with one command (or follow Steps 7 and 8 to use docker-compose instead):
sudo docker run -d -p 80:80 vulnerables/web-dvwa



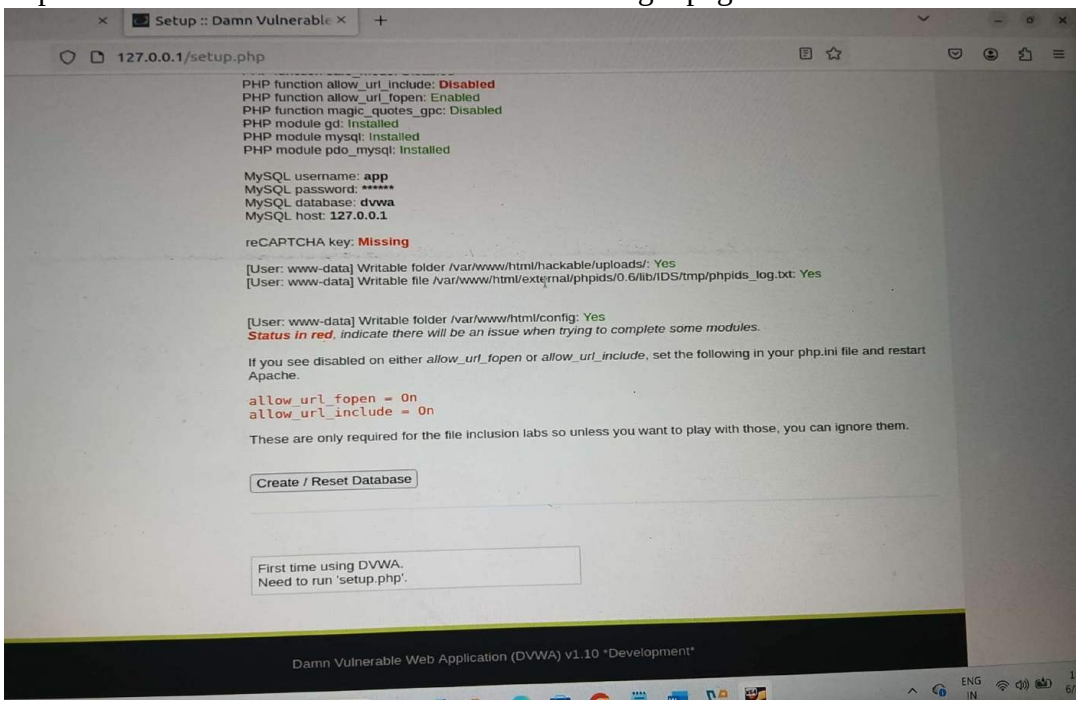


Step 7: Create the docker-compose file — Type: nano docker-compose.yml — then paste in the DVWA Docker configuration (available at github.com/digininja/DVWA). Press Ctrl+X, then Y, then Enter to save.

Step 8: Start DVWA — Type: sudo docker-compose up -d

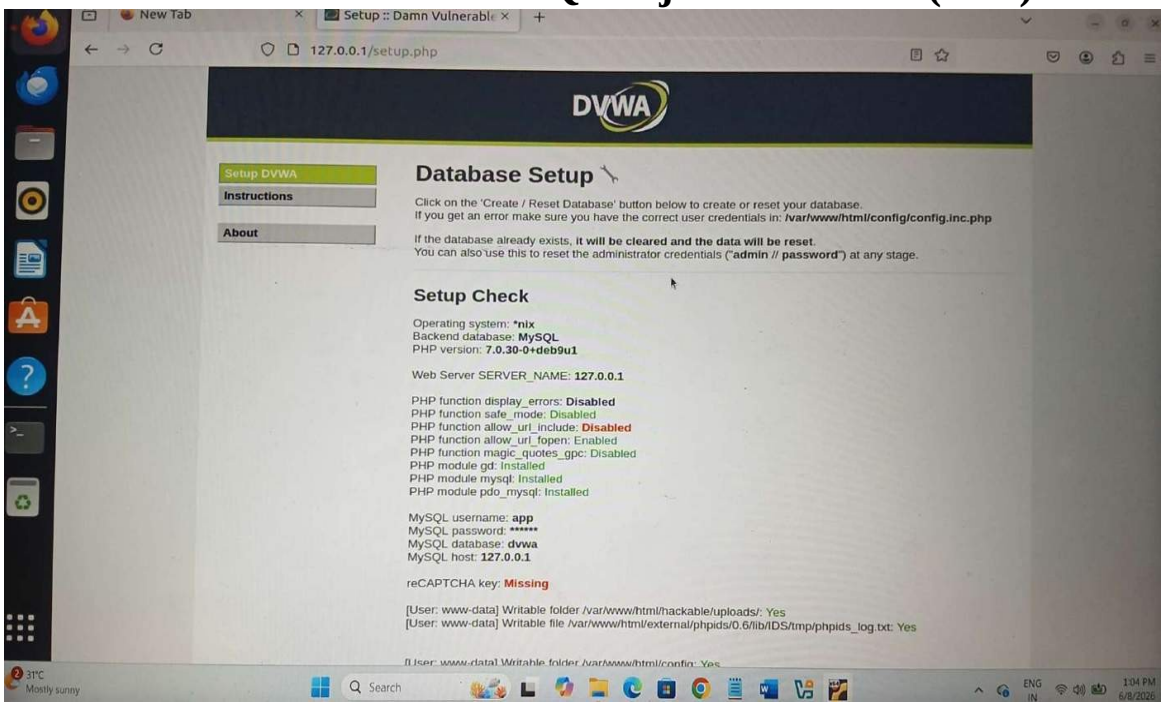

ubuntu [Running] - Oracle VirtualBox
Activities
Firefox Web Browser
Jun 8 13:06
New Tab
Welcome :: Damn Vulner
127.0.0.1/index.php
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript
DVWA Security
PHP Info
About
Welcome to Damn Vulnerable Web Application!
Damn Vulnerable Web Application (DVWA) is a PHP/MySQL web application that is damn vulnerable. Its main goal is to be an aid for security professionals to test their skills and tools in a legal environment, help web developers better understand the processes of securing web applications and to aid both students & teachers to learn about web application security in a controlled class room environment.
The aim of DVWA is to practice some of the most common web vulnerabilities, with various levels of difficultly, with a simple straightforward interface.
General Instructions
It is up to the user how they approach DVWA. Either by working through every module at a fixed level, or selecting any module and working up to reach the highest level they can before moving onto the next one. There is not a fixed object to complete a module; however users should feel that they have successfully exploited the system as best as they possible could by using that particular vulnerability.
Please note, there are both documented and undocumented vulnerability with this software. This is intentional. You are encouraged to try and discover as many issues as possible.
DVWA also includes a Web Application Firewall (WAF), PHPIDS, which can be enabled at any stage to further increase the difficulty. This will demonstrate how adding another layer of security may block certain malicious actions. Note, there are also various public methods at bypassing these protections (so this can be seen as an extension for more advanced users)!
There is a help button at the bottom of each page, which allows you to view hints & tips for that vulnerability. There are also additional links for further background reading, which relates to that security issue.
WARNING!
Damn Vulnerable Web Application is damn vulnerable! Do not upload it to your hosting provider's public html folder or any Internet facing servers, as they will be compromised. It is recommend using a virtual machine (such as VirtualBox or VMware), which is set to NAT networking mode. Inside a guest machine, you can downloading and install XAMPP for the web server and database.
31°C
Mostly sunny
Search
ENG
IN

Step 9: Access DVWA — Open the Firefox browser in Ubuntu and go to http://localhost:80. You should see the DVWA login page.

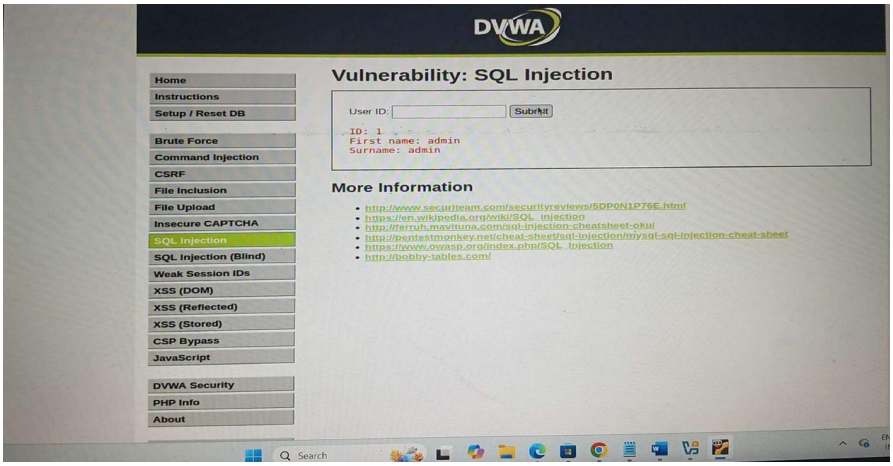


Step 10: Log in — Username: admin | Password: password. Click 'Create / Reset Database' on the setup page.

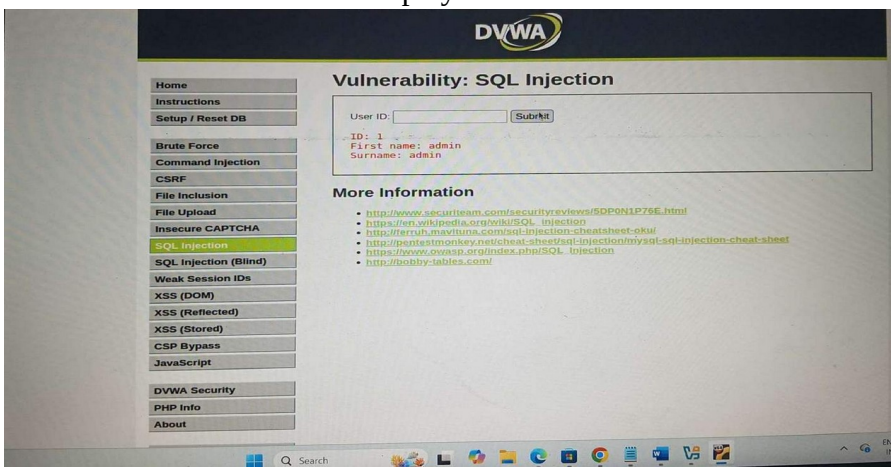
Phase 4 — Perform an SQL Injection Attack (Test)



This step simulates what an attacker does. We do this so you can see the attack in the logs — and then block it.

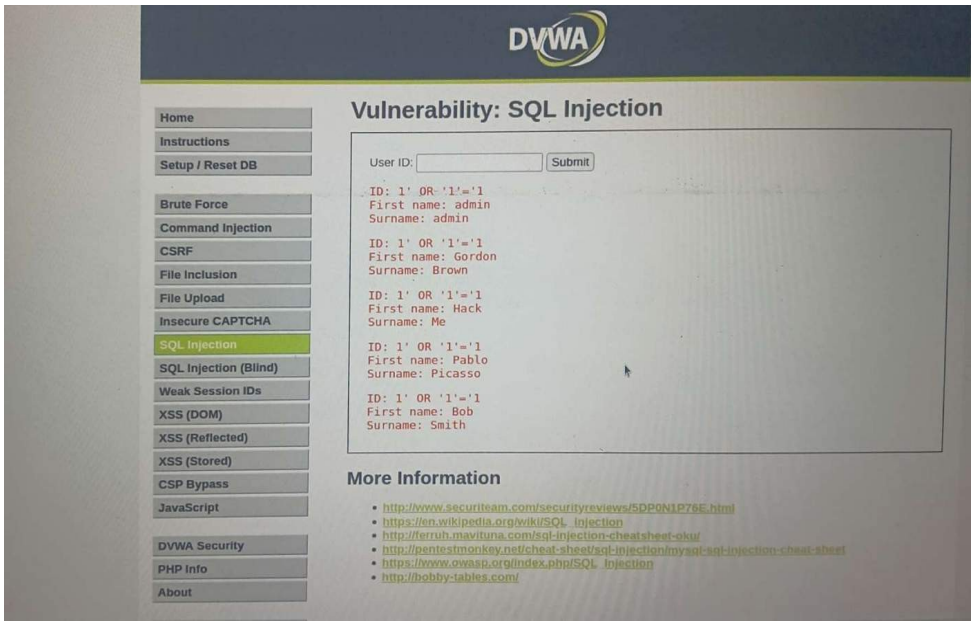


Step 11: Enter a normal input — In the User ID box, type 1 and click Submit. You should see the user's details displayed.



Step 12: Enter a malicious input — Now type the following into the User ID box and click Submit:

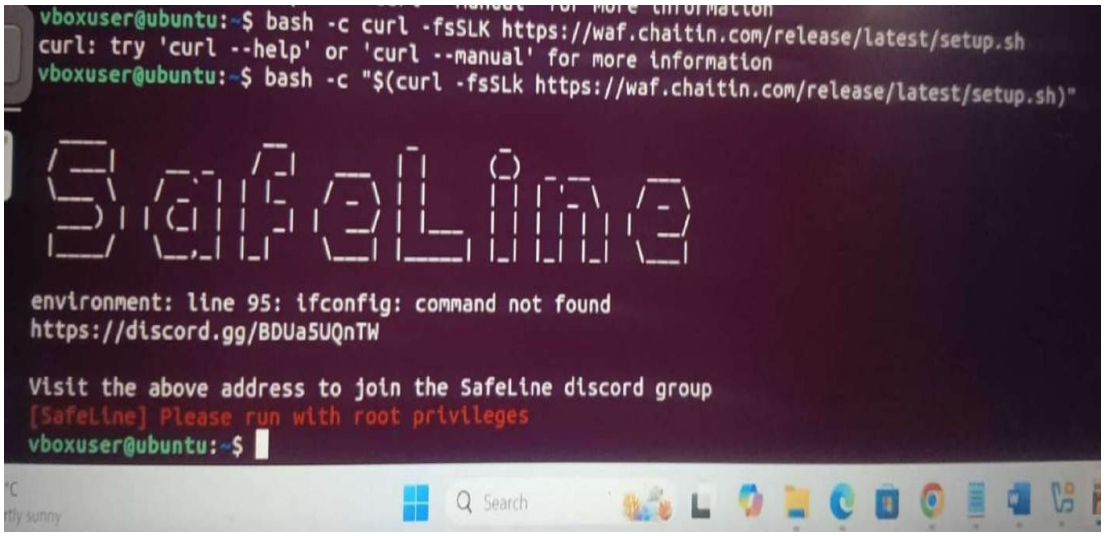
1' OR '1'='1



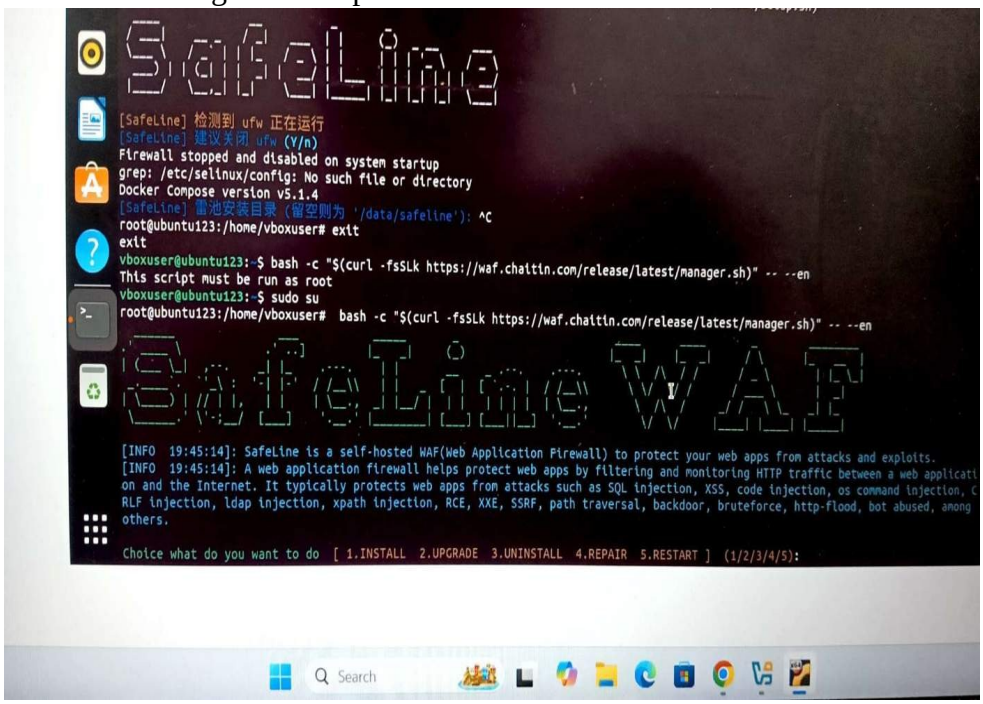
Phase 5 — Install SafeLine WAF

Step 13: Install SafeLine — In the terminal, run:

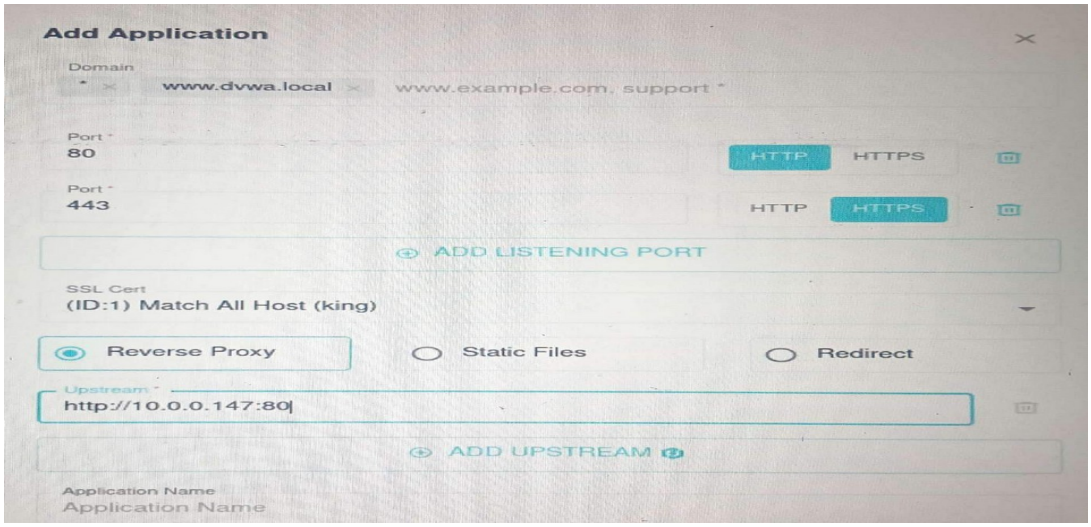
bash -c "$(curl -fsSLk https://waf.chaitin.com/release/latest/setup.sh)"

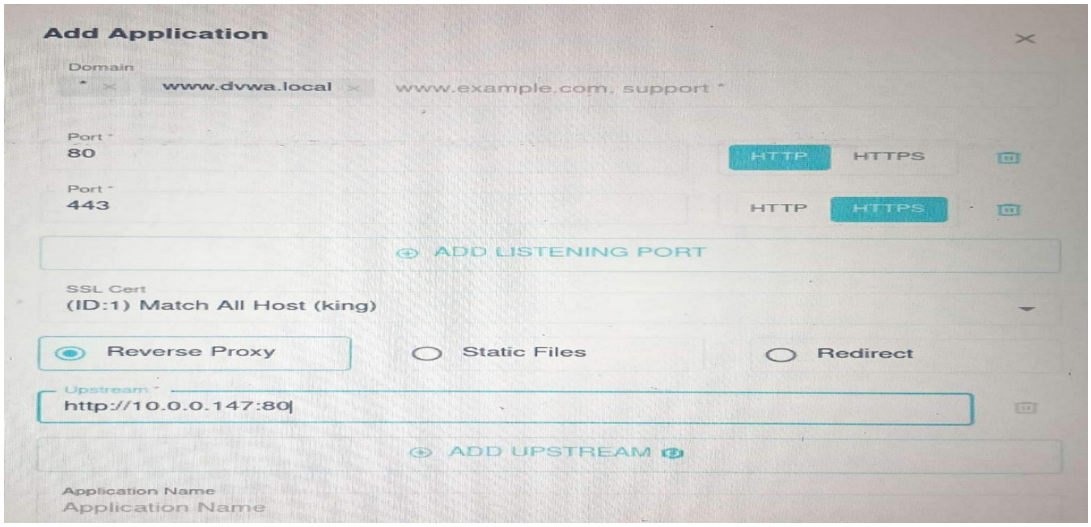

vboxuser@ubuntu:~$ bash -c curl -fsSLK https://waf.chaitin.com/release/latest/setup.sh
curl: try 'curl --help' or 'curl --manual' for more information
vboxuser@ubuntu:~$ bash -c "$(curl -fsSLk https://waf.chaitin.com/release/latest/setup.sh)"
environment: line 95: ifconfig: command not found
https://discord.gg/BDUa5UQnTW
Visit the above address to join the SafeLine discord group
[SafeLine] Please run with root privileges
vboxuser@ubuntu:~$
Search

Step 14: Follow the prompts — The installer will guide you through the setup. Accept all defaults for a beginner setup.



Step 15: Add your site — In the SafeLine dashboard, click 'Add Site' and configure it to protect your DVWA site running on port 80.



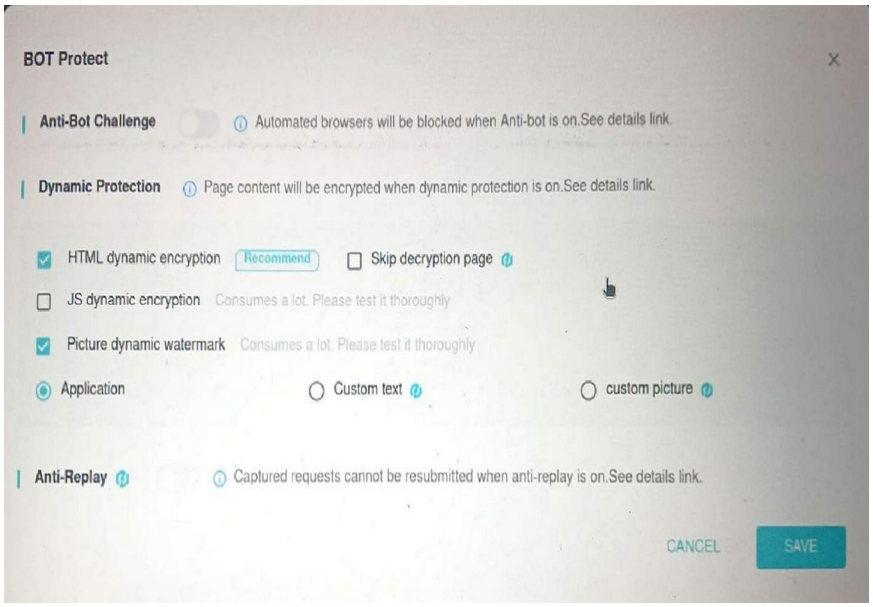


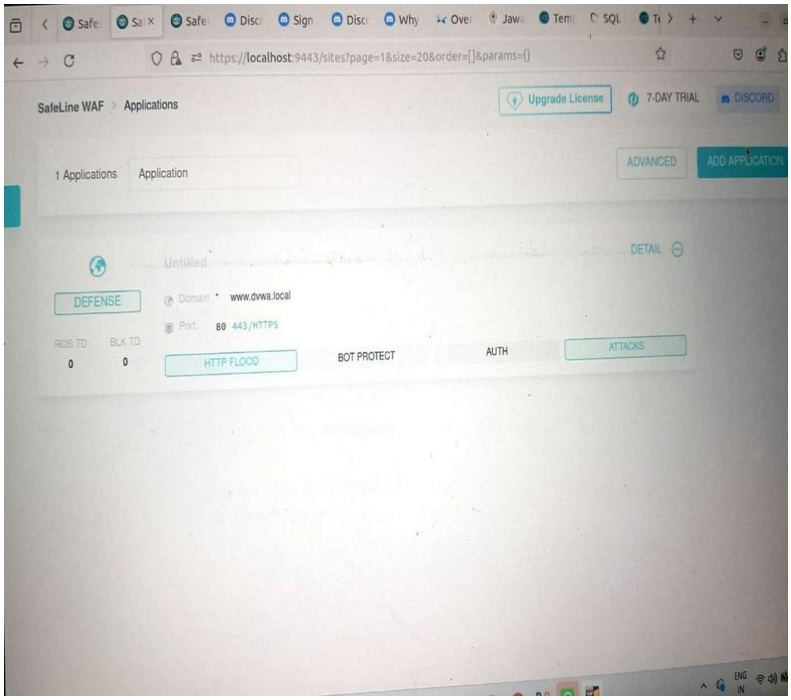
Step 16: Test the WAF — Try the SQL injection from Phase 4 again. This time, SafeLine should block it and you will see a block page instead of results.



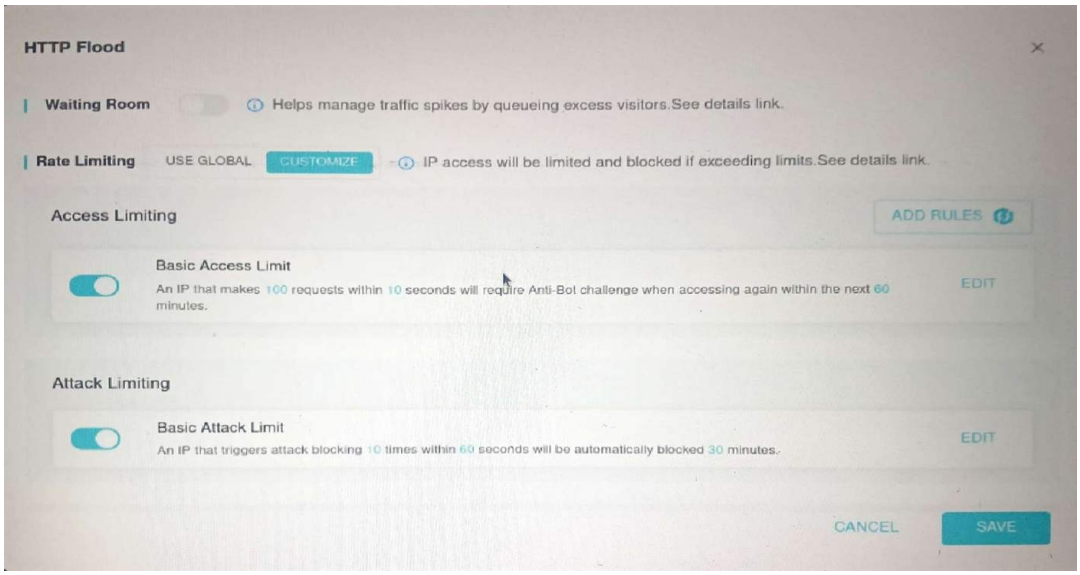
Phase 6 — Configure Rate Limiting

Step 17: Open the Rate Limiting settings — In the SafeLine dashboard, go to the 'Rate Limiting' or 'Anti-Bot' settings.

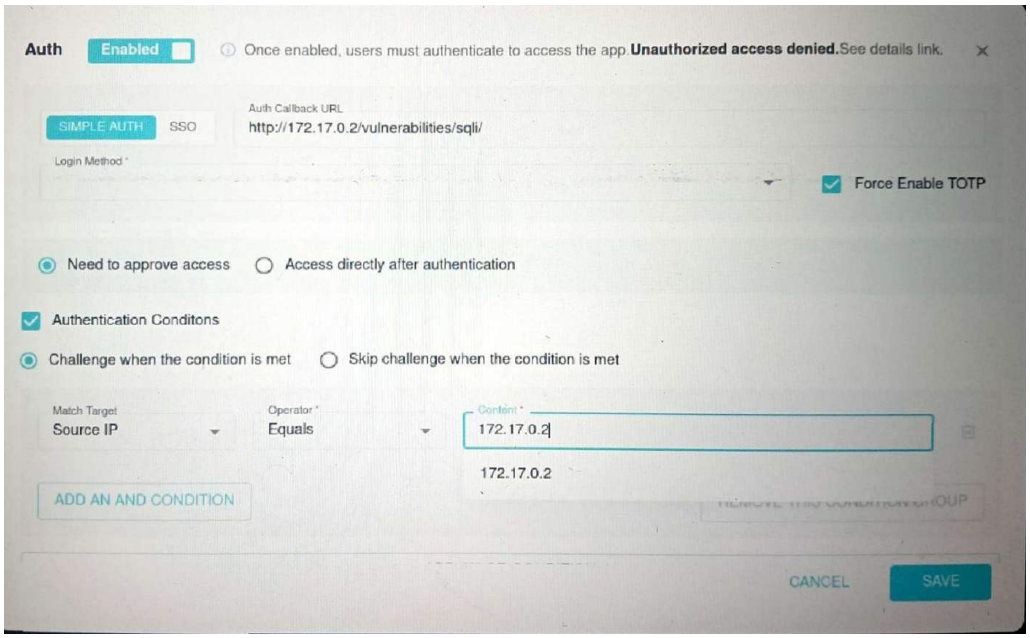


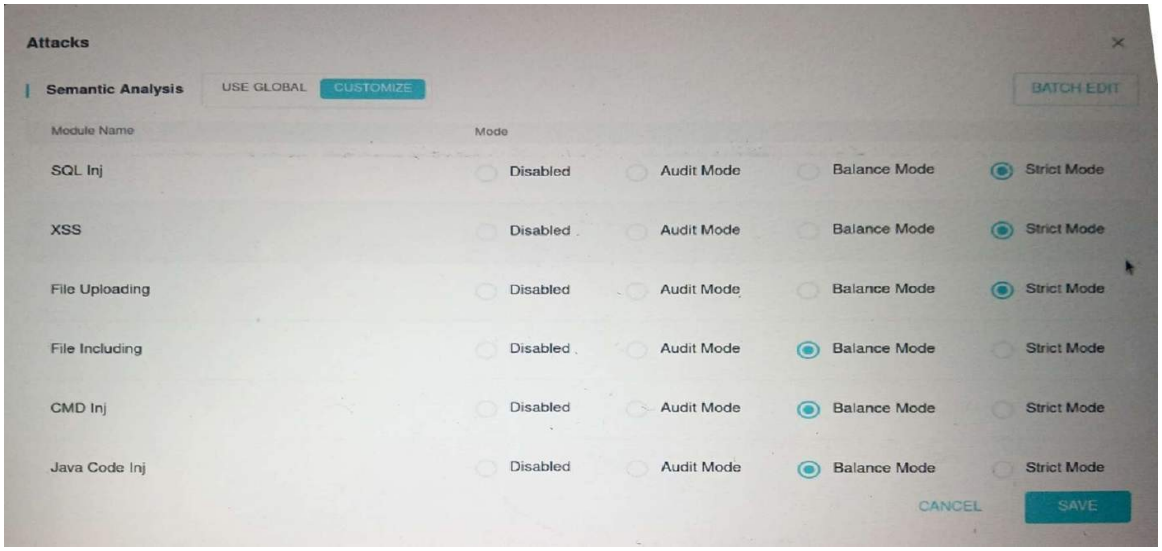


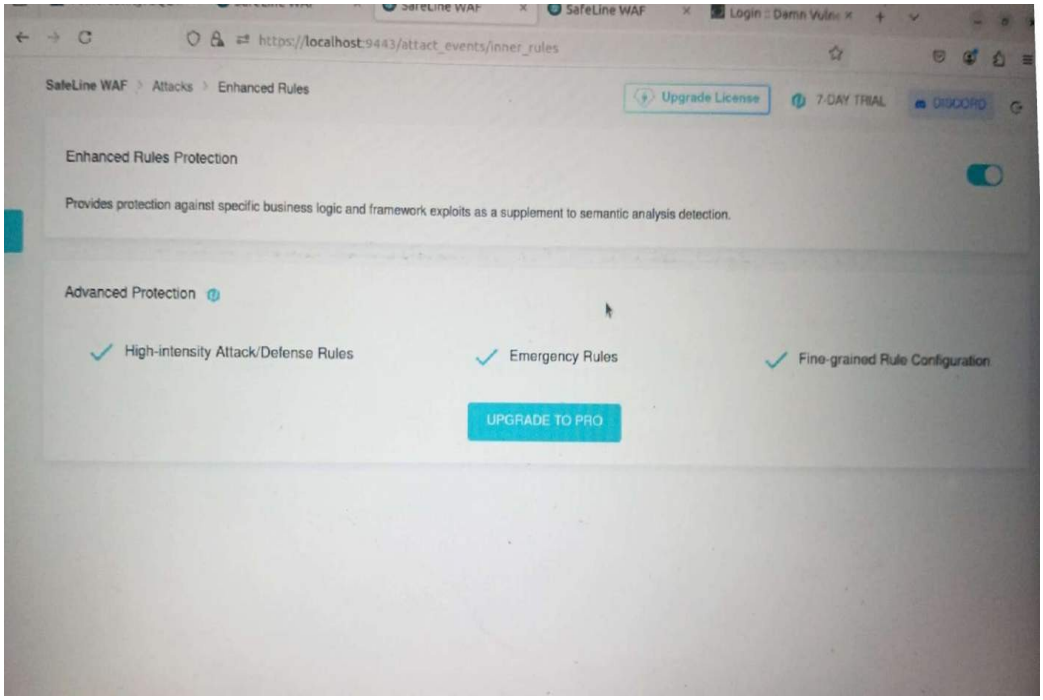
Step 18: Set a limit — Configure a rule that allows no more than 20 requests per second from a single IP address.



Step 19: Test it — You can simulate rapid requests by refreshing the DVWA page many times quickly, or by using a basic loop command in the terminal.


Auth
Enabled
Once enabled, users must authenticate to access the app Unauthorized access denied.See details link.
SIMPLE AUTH
SSO
Auth Callback URL
http://172.17.0.2/vulnerabilities/sqli/
Login Method *
Force Enable TOTP
Need to approve access
Access directly after authentication
Authentication Conditons
Challenge when the condition is met
Skip challenge when the condition is met
Match Target
Source IP
Operator *
Equals
Content *
172.17.0.2
172.17.0.2
ADD AN AND CONDITION
CANCEL
SAVE


Attacks
Semantic Analysis
USE GLOBAL
CUSTOMIZE
BATCH EDIT
Module Name
Mode
SQL Inj
Disabled
Audit Mode
Balance Mode
Strict Mode
XSS
Disabled
Audit Mode
Balance Mode
Strict Mode
File Uploading
Disabled
Audit Mode
Balance Mode
Strict Mode
File Including
Disabled
Audit Mode
Balance Mode
Strict Mode
CMD Inj
Disabled
Audit Mode
Balance Mode
Strict Mode
Java Code Inj
Disabled
Audit Mode
Balance Mode
Strict Mode
CANCEL
SAVE



How to Read the WAF Logs

In the SafeLine dashboard, navigate to the 'Events' or 'Logs' section. You will see entries showing:

• The IP address of the attacker

• The time of the attack

• What type of attack was detected (e.g., SQL Injection)

• Whether it was blocked or allowed

What Success Looks Like

• DVWA loads and you can log in

• SQL injection works when the WAF is NOT enabled (it returns all users)

• SQL injection is BLOCKED when the WAF IS enabled (an error or block page is shown)

• The block event appears in SafeLine logs

Conclusion:

This project demonstrates how SQL injection can be used to attack a website and how SafeLine WAF can prevent such attacks. First, a vulnerable website (DVWA) was deployed and tested with an SQL injection attack. SafeLine WAF was then installed and configured to protect the site. After the WAF was enabled, the attack was detected and blocked. This exercise provided practical knowledge of network security, common cyberattacks, and the importance of using WAFs to protect web applications.